LifeLink

A Secure Infrastructure-Independent Emergency Communication System

D1 Project Proposal and Initial Literature Review

TED University | Department of Computer Engineering | CMPE 491
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Project information	Details
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Project description

LifeLink is a proposed emergency communication system for situations in which mobile networks and internet access are unavailable. Earthquake rescue is the primary design and evaluation scenario. The system combines a low-power wearable for help requests and pulse indications, a separate offline messaging channel, and a local gateway with a rescue-team dashboard. Communication relies on the project's powered local equipment and an available radio path. The messaging interface and radio technology will be selected by the team after feasibility and cost checks.

The wearable will attempt pulse measurements after manual or rule-based motion activation and report whether the measurement is usable or uncertain. A help request will not depend on successful pulse measurement. The gateway will verify messages, manage excessive traffic, display measurement freshness, and relay an acknowledgement actually issued by a rescue operator. A small machine-learning model will be investigated for measurement-quality assessment and later on-device use as TinyML. LifeLink will support independent rescue work; it will not diagnose a person's condition or interpret silence as evidence that no living person is present.

Proposal objective

We aim to develop and evaluate a small end-to-end prototype that combines help communication, uncertainty-aware pulse reporting and bounded traffic control. The project brings together embedded systems, cybersecurity and machine learning. We will begin with a small demonstrator and use the results to guide later hardware and protocol decisions.

1 Problem and proposed system

Problem and intended users

During the earthquakes of 6 February 2023, people trapped under rubble used social media to request help [11]. At the same time, communication was disrupted: ITU describes power and fuel-supply problems affecting telecommunications [10], while OONI documented restrictions on access to Twitter on 8 February [12]. Network disruption and platform access restrictions are separate problems, but both matter when an urgent help request depends on an online service.

Some people helped collect help requests shared on social media and tried to relay them to rescue teams. We want to explore a local communication path for such requests. Automatic wearable reports can also support people unable to use a phone or press a button.

Component	Planned responsibility
Wearable	Manual help button, motion-based activation, periodic optical pulse acquisition, measurement-quality status, device-presence reports and user feedback. A removable capsule and interchangeable soft wrist strap are the initial form concept.
Offline messaging channel	Short help messages and operator replies without mobile internet. The team will compare a local Wi-Fi browser page with a radio-based interface before choosing the implementation.
Gateway and dashboard	Collect and verify messages; show wearable and messaging sessions; distinguish request, receipt and operator acknowledgement; display measurement age and last contact; apply bounded traffic controls.
Field relay nodes	Possible second-semester extension for collection and forwarding. They are not required for the initial direct-to-gateway demonstration.

Core interaction

The wearable wakes through a help-button action or a controlled motion trigger. It attempts a pulse measurement and sends a quality-labelled report. If pulse acquisition fails, it can still report device presence. An operator can acknowledge a help request through the dashboard. The wearable indicates that acknowledgement using vibration, with an LED as a secondary channel. The interface will distinguish a request created on the device, receipt at the gateway and acknowledgement by an operator.

In the messaging flow, a person sends a short message through the selected local interface, the message appears on the shared dashboard, and an operator reply returns through that interface. If the Wi-Fi option is selected, a gateway-hosted browser page is the proposed first demonstration; no installed phone application or field node would be needed for that option.

Design boundaries

The dashboard will show three main results: a pulse indication, a detected device with an uncertain measurement, or a missing expected response from a previously observed device. It will show measurement age and planned reporting pauses separately. Precise location estimation, respiration radar, room imaging, medical diagnosis and rescue-priority decisions are outside the initial scope.

2 Initial literature review

We reviewed work on disaster communication, wearable rescue transmitters, survivor detection and optical signal quality. We also considered an existing off-grid messaging platform and a standard for authenticated encryption. These sources guide our initial choices and the questions we will test.

2.1 Communication after an earthquake

Pueyo-Centelles and colleagues propose a LoRa-based system through which citizens can report their status to emergency organizations using predefined messages and a simple interaction [1]. For LifeLink, predefined reports offer a simple starting point for the wearable's help button. A separate interface can support more detailed messages.

Our radio comparison will consider receive windows, airtime and listening energy. It must also allow an operator to acknowledge a request after the original transmission window has closed.

2.2 An existing off-grid platform

Meshtastic is an open-source platform that uses LoRa radios for off-grid mesh communication and text exchange without relying on existing cellular or internet infrastructure [2]. We will use it as a practical reference when comparing messaging and radio options.

We are particularly interested in how a messaging platform could support pulse reports, measurement timestamps and operator acknowledgements. D2 will compare the candidate platforms and examine how they handle delivery and excessive traffic.

2.3 Wrist PPG and reference data

Reiss and colleagues present Deep PPG, a convolutional approach to heart-rate estimation, and introduce PPG-DaLiA [3]. The dataset provides recordings from 15 participants during daily activities; its wrist photoplethysmography (PPG) and acceleration recordings are paired with an electrocardiogram (ECG)-based reference [4]. It offers a suitable starting point for studying movement-related errors before collecting our own data.

Our model will assess the reliability of a fixed pulse-estimation method. We will label each window by comparing its estimate with the ECG-derived reference, then train a model using PPG and motion features. ECG will be used only for labels and evaluation. Training and test data will be separated by participant.

2.4 Wearable rescue transmitters

Avalanche transceivers provide a commercial example of body-carried rescue signalling. Mammut's Barryvox documentation describes an upper-body carrying system and a 457 kHz transmitter [7]. For our wearable, this example highlights the importance of secure attachment, simple controls and having the device ready when needed.

2 Initial literature review continued

2.5 Survivor detection beneath rubble

Rescue teams also use methods that do not require a transmitter on the affected person. NASA JPL describes FINDER radar sensing small motions associated with breathing and heartbeat beneath rubble [8]; Savox describes Delsar seismic/acoustic listening for trapped victims [9]. These examples help define LifeLink's role: the wearable supplies device reports and help requests, while rescue teams continue their independent search.

2.6 Signal quality features

Elgendi investigates several PPG signal-quality indices, including skewness and relative power [5]. This work motivates testing simple waveform and spectral features in our first quality model.

Our initial comparison will use rules, logistic regression and a small multilayer perceptron. Candidate features describe PPG peak power, a competing peak, motion variability, PPG–motion frequency separation and waveform skewness. We will compare them using the same pulse estimator and labels, looking at both accepted measurements and errors.

2.7 Authentication and confidentiality

NIST SP 800-38C defines CCM, which combines confidentiality and message authentication for a symmetric block cipher [6]. AES-CCM is therefore a candidate for protecting the wearable's reports and commands. The protocol design will also specify device keys, unique nonces, counters, associated data and recovery after a restart. Traffic limits will be handled separately.

Users will not need a name or account to send wearable reports. Authenticated temporary identifiers will let the gateway manage device sessions; the gateway may still link successive reports from the same device. The first prototype will keep personal medical profiles and continuous location tracking outside its scope.

Proposed contribution and scope

Our contribution is to bring help messaging, pulse-quality reporting, clear acknowledgement states and traffic control into one prototype. We will evaluate how these functions work together within the hardware, energy and airtime budget.

Published results from daily-activity PPG, avalanche transmitters and rescue radar provide useful starting points, but our selected sensors and radio link need their own tests. Communication requires powered local equipment and a usable link. Independent rescue search remains necessary for people without a device or a working connection, and silence will never be interpreted as absence of life.

Literature direction	How it informs LifeLink
Earthquake status messaging [1]	Predefined help reports and a simple civilian interaction.
Off-grid mesh platform [2]	A practical communication reference; adoption remains open.
Body-carried rescue transmitter [7]	A commercial reference for wearing and readiness; different propagation conditions.
Survivor detection beneath rubble [8, 9]	Radar and seismic/acoustic search without a carried transmitter; LifeLink complements independent rescue search.
Deep PPG and PPG-DaLiA [3, 4]	Motion-aware data and ECG reference for participant-separated evaluation.
PPG quality indices [5]	Small feature-based models and explicit quality baselines.
CCM standard [6]	A candidate basis for report and command protection.

3 Scope work plan and evaluation

Period	Planned output
Semester one	Requirements, analysis and high-level design reports; a small wearable–gateway flow and a separate offline messaging flow on one dashboard; rule-based measurement quality; authenticated temporary identifiers; a bounded routine-report restriction; an initial computer-based quality model.
Semester two	Deploy the feasible model on the chosen MCU; measure the full processing pipeline; expand range/obstruction and multi-device load tests; investigate automatic traffic control and field relays if feasible.

Initial demonstration

Flow A begins with controlled activation of the wearable. A pulse indication or an uncertain measurement is reported to the dashboard. An operator acknowledges a request, and the wearable gives feedback. A timed restriction pauses routine reports while manual help and scheduled control listening remain available within their traffic budgets. Routine reporting resumes when the restriction expires.

In Flow B, a short help message reaches the same dashboard and an operator reply returns through the selected interface. Both flows will be demonstrated with internet access disabled. The team will finalise the channel details.

Evaluation approach

Communication tests will record message delivery, delay and acknowledgement status. Traffic-quarantine tests will compare the same load with the mechanism enabled and disabled, including delayed/lost help messages and false restrictions. Authentication tests will include modified or replayed commands. Power tests will measure the complete prototype board.

For quality assessment, a fixed spectral pulse extractor will be evaluated against ECG-derived reference labels. Models and rules will be compared at similar coverage, reporting errors among accepted windows and results by participant and activity. The initial model candidates are logistic regression and a small single-hidden-layer MLP. In semester two, we will deploy the selected approach and measure its memory use, latency and energy consumption.

Feasibility and responsibilities

MAX30101 and a low-power accelerometer are sensor candidates. MCU, radio, battery chemistry and capacity remain subject to laboratory inventory, staged purchasing and measured feasibility. We will assess range and service life through testing. Wearable comfort, battery protection and maintenance will also guide the design. Human measurements will follow the applicable university ethics process. We can begin with open data and hardware tests that do not involve participants.

The embedded-systems role covers sensing, activation, power and feedback; cybersecurity covers authentication, temporary identifiers and traffic control; AI covers quality rules, labels and model evaluation. Integration, messaging-interface selection and reports are shared work. Individual task assignment will be agreed by the team.

4 Milestones and references

Deadline	Deliverable
16 October 2026	D1 Project proposal and initial literature review.
23 October 2026	D2 Project specifications and literature review; D3 backlog.
20 November 2026	D4 Analysis report; D5 updated backlog.
31 December 2026	D6 High-level design report; D7 updated backlog.
4–17 January 2027 TBA	D8 Presentation, poster and initial demonstration.

Dates follow the semester-specific schedule in the CMPE 491 syllabus. The team will record its interface, radio and component decisions as the feasibility work progresses.

References

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